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CRM Assistant

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VOLUNTEER ENGAGEMENT

01 The problem

Staff spend a large share of their day on repetitive work: looking up student records, writing routine emails, and updating fields by hand. Information about any one student is spread across records, notes, and documents, so answering even a simple question means digging through the system manually. This is slow, and it means students who need attention can slip through unnoticed.

02 The solution

An assistant built into the CRM that staff operate in plain English. A staff member types what they need, and the assistant retrieves the right records, summarises a student's history, drafts a message, flags students at risk of dropping out, or updates a record on request. It runs entirely on local hardware, so student data never leaves the office.

03 Objectives

- Cut the time staff spend on manual lookups and routine writing
- Surface at-risk students and cooling leads earlier, before they are lost
- Keep all student data on the organisation's own machines
- Stay reliable and accurate enough to be trusted in daily use

04 Who it is for

Front-desk, admin, and student-support staff. The people using it are not technical, so every interaction happens in plain language with no query syntax or forms to learn.

05 Scope

In scope

- Five staff actions: query, summarise, draft, flag, update
- A small predictive model for at-risk detection
- An evaluation harness to measure accuracy
- A realistic mock CRM database to build against

Out of scope (for now)

- Live production data on day one
- Sending emails automatically without human review
- Document scanning and image reading
- Changes to the existing CRM's core software

06 Success criteria

- Common lookups can be completed in a single plain-English request
- At-risk flags line up with what actually happened in past records
- Measured accuracy on a fixed test set stays above an agreed threshold
- The whole system runs comfortably within 24GB of memory on local hardware

07 Phased plan

PHASE 1 **Retrieval and Q&A**

Staff can ask questions and get answers pulled from records and policy documents.

PHASE 2 **Actions and predictions**

Add summarising, email drafting, record updates, and the at-risk model.

PHASE 3 **Evaluation and hardening**

Build the accuracy test harness and tighten reliability before wider use.

08 Technology

Python FastAPI ChromaDB sentence-transformers

Qwen3-14B (LM Studio / Ollama) FastMCP SQLite scikit-learn
